COURSE SLIDES

Certified Lean Six Sigma Black Belt (CLSSBB) Training

WSQ Course Code: TGS-2024051900
Conducted by Tertiary Infotech Academy Pte Ltd

Version 1

ADMIN

Welcome and Course Administration

- Training hours: 9:00 AM to 6:00 PM each day.
- Please sign in and complete SSG digital attendance as instructed.
- Keep your lab journal open throughout the course.

About the Trainer

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Your Trainer

General Trainer template -
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

YOUR TRAINER

About the Trainer

AA

Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

QUALIFICATIONS

PhD - specialises in AI, automation and software engineering.

DELIVERS

WSQ courses on Lean Six Sigma, analytics, AI automation and app development.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Digital Attendance is Mandatory

- AM, PM, and assessment attendance must be completed where required.
- Minimum attendance requirements apply for funding and assessment eligibility.
- Use the displayed QR code or portal instructions from the trainer.

HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively - no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

ADMIN

Courseware and LMS

- Slides, learner guide, lab instructions, and assessment access are provided through the LMS.
- Save a local copy of your project artifacts after each lab.
- Use the lab files as the authority for hands-on tasks.

ASSESSMENT

Assessment Overview

- Final assessment and project review take place on Day 2.
- Assessment checks Black Belt concepts and application to DMAIC project artifacts.
- Open courseware where permitted by the trainer's assessment instructions.

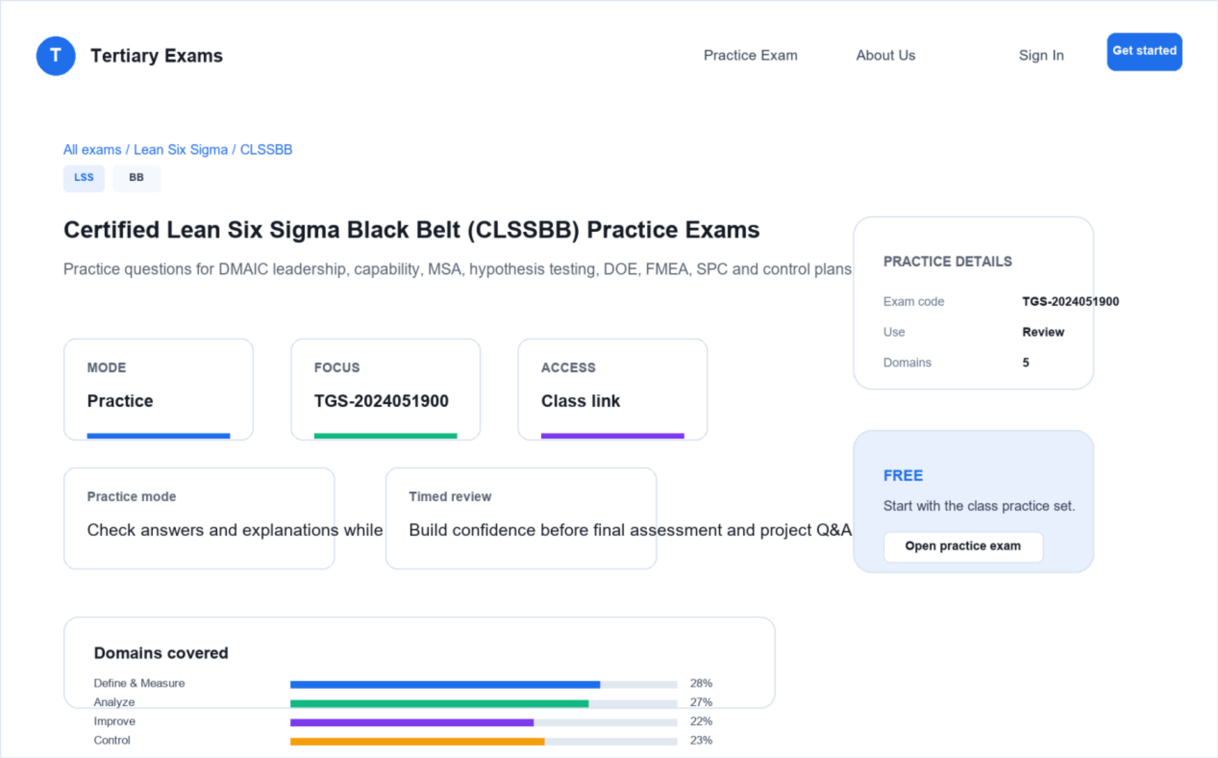
Assessment Flow Reminder

1. TRAQOM digital attendance
2. Assessment digital attendance
3. Sit WA then Case Study
4. Submit answers on the LMS
5. Sign Assessment Summary Record

The TRAQOM QR code and LMS submission are both handled through the LMS/TMS portal.

Take the Online CLSSBB Practice Exams

- Reinforce all five DMAIC phases with realistic, exam-style questions before assessment day.
- Use practice sets to test capability, MSA, hypothesis testing, DOE, FMEA, SPC and control plans.
- Review mode helps you check the correct answer and revisit the concept after each question.
- Complete the practice questions from Google Classroom, then bring difficult items to the review session.



Access and practise at:

Google Classroom > Practice Exams

COURSE OUTCOMES

Learning Outcomes

- Lead a DMAIC improvement project at Black Belt level.
- Use statistical and Lean tools to validate decisions.
- Build deliverables that transfer ownership to the process owner.

COURSE MAP

Course Flow

- Day 1: Define and Measure phases.
- Day 2: Analyze, Improve, Control, assessment and closure.
- All labs use one project scenario end to end.

Tools Used

- Excel, LibreOffice Calc, or Google Sheets for analysis.
- diagrams.net or equivalent for maps and fishbone diagrams.
- Project templates for charter, SIPOC, FMEA, control plan, and storyboard.

LABS

Lab Rules

- Use one project scenario across all labs.
- Keep formulas visible where possible.
- Do not claim root cause without evidence.
- Save every chart and artifact for the final project storyboard.

Version Control and Copyright

- Version 1.0 issued 9 July 2026.
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COURSE MAP

DMAIC is the spine of the course

- Each lab produces an artifact reused later.
- Define and Measure establish trustworthy scope and baseline.
- Analyze, Improve, and Control convert evidence into sustained results.

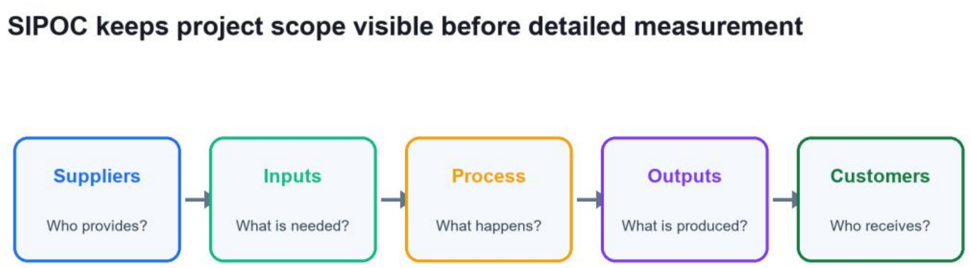
PROJECT STORY

Black Belt project artifacts accumulate across the labs

- Charter and CTQs define what matters.
- SIPOC, maps, and data plans define how to measure.
- Tests, FMEA, and control plans define how to decide and sustain.

SIPOC links customer requirements to the process boundary

- Start broad enough to see suppliers and customers.
- Use the map to choose useful data collection points.
- Update scope notes when the map exposes ambiguity.



Measurement quality comes before statistical confidence

- Operational definitions prevent inconsistent data.
- MSA risk notes document whether the metric can be trusted.
- Capability claims require stable and relevant data.

ANALYZE

Root causes need an evidence chain

- Use fishbone and 5 Whys to generate causes.
- Use Pareto, tests, regression, or ANOVA to validate causes.
- Separate statistical significance from practical significance.

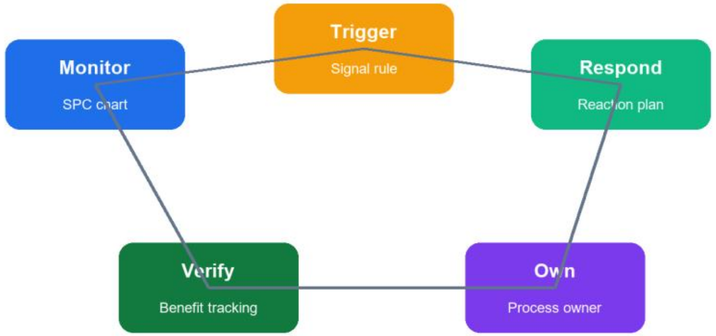
Evidence chain from baseline to validated root cause

Baseline data	Clean data, capability, charts
Variation view	Run chart and control chart
Suspected causes	Fishbone, 5 Whys, Pareto
Statistical test	Hypothesis, regression, ANOVA
Decision	Validated, rejected, or more data

Control turns improvement into ownership

- SPC detects performance drift.
- Response plans tell owners what to do.
- Benefits tracking keeps gains visible after handover.

Control phase turns a pilot into sustained performance



LAB OVERVIEW

Lab 01: Project Charter, VOC, and CTQ Tree

- Select a Black Belt project scenario.
- Build a project charter.
- Capture voice of customer.
- Translate VOC into measurable CTQs.

STEP-BY-STEP

Lab 01 scenario and work sequence

- You have been assigned to lead a process improvement project. Before collecting data, you must define the business problem, scope, customers, and measurable success criteria.
- Select a process problem from your workplace or use a trainer-provided scenario.
- Write a one-paragraph business case.
- Write the problem statement using current performance facts.
- Avoid including assumed causes in the problem statement.
- Write a SMART goal statement.

VALIDATION

Lab 01 completion check

- The problem statement includes measurable current performance.
- The goal statement has target and timeline.
- Each CTQ can be measured.
- Scope boundaries are clear enough to prevent project drift.
- Deliverables: Project charter.; RACI table.; VOC table.; CTQ tree.

REFLECT

Lab 01 review questions

- Why should a problem statement avoid assumed causes?
- What makes a CTQ measurable?
- How does project scope protect the team?
- What is the Black Belt's role in stakeholder alignment?

LAB OVERVIEW

Lab 02: SIPOC, Process Map, and Waste Analysis

- Build a SIPOC.
- Map the current-state process.
- Identify handoffs, queues, rework, and waste.
- Refine the project scope.

STEP-BY-STEP

Lab 02 scenario and work sequence

- The project charter defines the problem, but the team needs a shared view of the process before measuring performance. You will map the current process and identify where waste may exist.
- Write the process start event and end event.
- List the main customers and outputs.
- List the major process steps between start and end.
- List key inputs and suppliers.
- Build a SIPOC table.

VALIDATION

Lab 02 completion check

- The SIPOC matches the project scope.
- The process map is understandable to someone outside the team.
- Waste examples are linked to actual steps.
- Data collection points are identified.
- Deliverables: SIPOC.; Current-state process map.; Waste analysis table.; Updated scope notes.

REFLECT

Lab 02 review questions

- Why is SIPOC useful before detailed process mapping?
- What is the difference between value-added and business-value-added work?
- How can process maps reveal hidden rework?
- Why should data collection points be chosen after mapping?

LAB OVERVIEW

Lab 03: Data Collection, MSA, and Process Capability

- Create a data collection plan.
- Define units, defects, and opportunities.
- Assess measurement system risks.
- Calculate baseline process performance.

STEP-BY-STEP

Lab 03 scenario and work sequence

- The team needs trustworthy baseline data. You must define the metric carefully, check whether measurement is reliable, and calculate current process capability.
- Select the primary Y metric from the CTQ tree.
- Write an operational definition for the metric.
- Define unit, defect, opportunity, target, and specification limits if available.
- Choose sampling period, sample size, data source, and data owner.
- Build a data collection plan.

VALIDATION

Lab 03 completion check

- The metric has an operational definition.
- Data collection responsibility is assigned.
- Measurement risks are documented.
- Capability calculations match the data type.
- Deliverables: Data collection plan.; Operational definition.; MSA risk notes.; Capability worksheet.

REFLECT

Lab 03 review questions

- Why can poor measurement systems invalidate a project?
- When would you use DPMO instead of Cp/Cpk?
- What is the difference between short-term and long-term capability?
- Why must specification limits come from customer or business requirements?

LAB OVERVIEW

Lab 04: Baseline Analysis and Control Charts

- Summarize baseline data.
- Create visual analysis charts.
- Select appropriate control charts.
- Interpret stability and variation.

STEP-BY-STEP

Lab 04 scenario and work sequence

- Before searching for root causes, the team must understand baseline variation. A process that is unstable should not be summarized only by averages.
- Import baseline data into a spreadsheet.
- Check for missing values, duplicates, and obvious data entry errors.
- Calculate count, mean, median, standard deviation, minimum, maximum, and range.
- Create a histogram.
- Create a time series or run chart.

Lab 04 completion check

- Descriptive statistics are calculated correctly.
- The chart type matches the data.
- Special cause signals are clearly marked.
- The baseline interpretation separates average performance from variation.
- Deliverables: Descriptive statistics table.; Histogram.; Run chart or time series plot.; Control chart.; Baseline interpretation.

REFLECT

Lab 04 review questions

- Why should stability be checked before capability claims?
- What is common cause variation?
- What is special cause variation?
- How does chart selection depend on data type?

LAB OVERVIEW

Lab 05: Root Cause Analysis and Hypothesis Testing

- Generate possible root causes.
- Prioritize causes using data.
- Select appropriate hypothesis tests.
- Interpret statistical and practical significance.

STEP-BY-STEP

Lab 05 scenario and work sequence

- The team has many opinions about why the process performs poorly. Your role is to separate assumptions from validated root causes.
- Create a fishbone diagram for the project problem.
- Use categories such as people, process, equipment, material, measurement, environment, and management.
- Select at least two likely causes and apply 5 Whys.
- Create a Pareto chart if category defect data is available.
- Translate suspected causes into testable hypotheses.

VALIDATION

Lab 05 completion check

- Each tested cause has a clear hypothesis.
- Test choice matches the data situation.
- Conclusions include practical significance.
- Unsupported causes are not treated as proven.
- Deliverables: Fishbone diagram.; 5 Whys worksheet.; Pareto chart if applicable.; Hypothesis test log.; Validated root cause list.

REFLECT

Lab 05 review questions

- Why is a low p-value not enough by itself?
- What is the difference between alpha and beta risk?
- When would you choose a non-parametric test?
- Why should root cause validation happen before solution design?

LAB OVERVIEW

Lab 06: Regression, ANOVA, and Design of Experiments

- Study relationships between X variables and Y performance.
- Interpret regression and ANOVA output.
- Plan a designed experiment.
- Identify experiment risks and controls.

STEP-BY-STEP

Lab 06 scenario and work sequence

- The team has narrowed the likely drivers. You will use analytical methods to understand relationships and plan an experiment to test improvement settings.
- Select two or three suspected X variables from Lab 05.
- Create scatter plots for continuous X and continuous Y relationships.
- Calculate correlation where appropriate.
- Run simple linear regression for one likely driver.
- Interpret slope, intercept, R-squared, p-value, and residuals.

Lab 06 completion check

- Regression or ANOVA interpretation is tied to the project question.
- Assumptions and limitations are documented.
- DOE factors and levels are controllable.
- Experiment risks are identified before execution.
- Deliverables: Scatter plot or group comparison chart.; Regression or ANOVA summary.; DOE planning worksheet.; Experiment risk notes.

REFLECT

Lab 06 review questions

- What does R-squared explain?
- Why should residuals be checked?
- What question does ANOVA answer?
- Why are randomization and replication important in DOE?

Lab 07: Improvement Selection, FMEA, and Pilot Plan

- Generate improvements linked to validated causes.
- Select solutions using a decision matrix.
- Assess risk with FMEA.
- Plan a controlled pilot.

STEP-BY-STEP

Lab 07 scenario and work sequence

- The team is ready to improve the process. The challenge is to choose solutions that address proven causes and can be tested without creating new risks.
- List validated root causes from Lab 05 and Lab 06.
- Brainstorm improvement ideas for each validated cause.
- Remove ideas that are not linked to a validated cause.
- Create a solution selection matrix.
- Score each solution by impact, effort, cost, time, risk, and stakeholder acceptance.

Lab 07 completion check

- Selected improvements address validated causes.
- FMEA actions are assigned to owners.
- Pilot success criteria are measurable.
- Rollback conditions are clear.
- Deliverables: Solution selection matrix.; FMEA.; Pilot plan.; Stakeholder communication notes.

REFLECT

Lab 07 review questions

- Why should solution selection wait until causes are validated?
- What do severity, occurrence, and detection measure?
- Why is mistake-proofing often stronger than inspection?
- What makes a pilot safer than a full rollout?

LAB OVERVIEW

Lab 08: Control Plan, SPC, and Benefits Realization

- Create a control plan and response plan.
- Select ongoing SPC measures.
- Define ownership and handover.
- Track benefits and close the project.

STEP-BY-STEP

Lab 08 scenario and work sequence

- The pilot has shown improvement. You must ensure the process owner can sustain the gains and respond when performance drifts.
- List the improved process steps that require control.
- Identify critical inputs and outputs.
- Define specification limits, target values, and measurement method.
- Select monitoring frequency and sample size.
- Assign process owner and backup owner.

VALIDATION

Lab 08 completion check

- Control plan has clear owners and frequency.
- Response plan tells the process owner what to do when performance changes.
- Benefit claims are traceable to data.
- Final storyboard tells the DMAIC story clearly.
- Deliverables: Control plan.; Response plan.; SPC monitoring plan.; Benefits realization worksheet.; Final DMAIC storyboard.

REFLECT

Lab 08 review questions

- Why do improvements fail without a control plan?
- What is the difference between monitoring and response?
- Why should benefits be tracked after implementation?
- How does a final storyboard support project closure?

Certification and TRAQOM Survey

- Complete TRAQOM and course feedback requirements.
- Confirm all required attendance records are complete.
- Check certificate and competency instructions with the trainer.

CLOSING

Project Storyboard Readiness

- Tell the DMAIC story using evidence from each lab.
- Show how the selected improvement links to validated root causes.
- Explain the control plan and how benefits will be sustained.

CLOSING

Thank You

- You have completed the CLSSBB DMAIC course flow.
- Keep using evidence, operational definitions, and control plans in real projects.
- Questions and next steps.